

XUAN LIU

Scripps Institution of Oceanography, University of California San Diego
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EDUCATION

Washington University in St. Louis

Ph.D. Energy, Environmental & Chemical Engineering
• Supervisors: Prof. Jay Turner & Prof. Randall Martin
• GPA: 3.96/4.00

St. Louis, Missouri, USA
2019 – 2024

Xiamen University

B.S. Chemical Engineering and Technology
• Overall GPA: 3.86/4.00; Major GPA: 3.94/4.00
• Rank: 1/60

Xiamen, Fujian, China
2015 – 2019

RESEARCH EXPERIENCE

Postdoctoral Scholar at University of California San Diego, Advisor: Amato Evan

- Assess the impact of dust from ephemeral lakes on Earth's weather and climate using the newly developed soil mineralogy dataset from NASA's Earth surface Mineral dust source Investigation (EMIT) mission (2024 – present)
- Examine changes in exposure to playa dust and associated health risks as the Salton Sea continues to shrink using the Weather Research and Forecasting Model with Chemistry (WRF-Chem) (2025 – present)

Graduate Research Assistant at Washington University in St. Louis, Advisors: Jay Turner & Randall Martin

- Investigated the characteristics of airborne dust and trace elements in Uzbekistan with measurements from filter samples, low-cost sensors, and soil samples (2022 – 2024)
- Examined the attenuation effect due to particle size and mass loading for crustal elements in X-ray Fluorescence analysis of ambient particulate matter samples (2022 – 2024)
- Developed the methodology and explored implications for elemental analysis of ambient particulate matter samples from the global Surface PARTICulate mATter Network (SPARTAN) (2021 – 2023)
- Developed and evaluated a global-scale mineral dust equation with region-specific coefficients for estimating dust mass (2020 – 2022)

Undergraduate Research Assistant at Xiamen University, Advisors: Qiugen Zhang, Gaofeng Zheng, & Song Tu

- Developed a new type of side-chain-type polymer anion exchange membranes for fuel cells (2017 – 2018)
- Applied electrospinning to prepare polymer nanofibers for heavy metal ion adsorption (2016 – 2018)
- Investigated optimal purification conditions of 2-imidazolidinone (2016 – 2018)

PUBLICATIONS

- **Liu, X.**, Turner, J. R., Hand, J. L., Schichtel, B. A., Martin, R. V. (2022). A Global-Scale Mineral Dust Equation. *Journal of Geophysical Research: Atmospheres*, 127(18), e2022JD036937. <https://doi.org/10.1029/2022JD036937>
- **Liu, X.**, Turner, J. R., Oxford, C., McNeill, J., Walsh, B., Roy, E. L., ... Martin, R. V. (2024). Elemental Characterization of Ambient Particulate Matter for a Globally Distributed Monitoring Network: Methodology and Implications. *ACS EST Air*, 1(4), 283-293. <https://doi.org/10.1021/acsestair.3c00069>
- **Liu, X.**, Turner, J. R., Mitroo, D., Ren, Y., Oxford, C. R., Liu, W., & Martin, R. V. (2025). Assessing Attenuation Effects in X-ray Fluorescence Analysis of Light Elements in Mineral Dust. *ACS ES&T Air*, 3(1), 175-185. <https://doi.org/10.1021/acsestair.5c00295>
- **Liu, X.**, Amonov, M., & Turner, J. R. (2025). Quantifying Dust Contribution to Particulate Matter in Central Asia: Insights from the Elemental Composition of PM_{2.5} and PM₁₀ in Uzbekistan. *Atmospheric Pollution Research*, <https://doi.org/10.1016/j.apr.2025.102781>
- Zhang, D., Martin, R. V., **Liu, X.**, Van Donkelaar, A., Oxford, C. R., Li, Y., ... & Windwer, E. (2025). Improving annual fine mineral dust representation from the surface to the column in GEOS-Chem 14.4. 1. Geoscientific Model Development, 18(19), 6767-6803. <https://doi.org/10.5194/gmd-18-6767-2025>
- Eom, S., Park, S. S., **Liu, X.**, Oxford, C. R., Martin, R. V., Kim, J., & Song, C. K. (2025). Impact of Chemical

Composition on Aerosol Scattering: Insights from the Surface Particulate Matter Network and Aerosol Robotic Network. *Environmental Science & Technology*, 59(37), 19899-19909. <https://doi.org/10.1021/acs.est.5c09325>

- Ren, Y., Oxford, C. R., Zhang, D., **Liu, X.**, Zhu, H., Dillner, A. M., ... & Martin, R. V. (2025). Black carbon emissions generally underestimated in the global south as revealed by globally distributed measurements. *Nature communications*, 16(1), 7010. <https://doi.org/10.1038/s41467-025-62468-5>
- Li, C., Martin, R. V., van Donkelaar, A., Jimenez, J. L., Zhang, Q., Turner, J. R., **Liu, X.**, ... & Thurston, G. D. (2025). Estimates of submicron particulate matter (PM₁) concentrations for 1998–2022 across the contiguous USA: leveraging measurements of PM₁ with nationwide PM_{2.5} component data. *The Lancet Planetary Health*, 9(6), e491-e502. [https://doi.org/10.1016/S2542-5196\(25\)00094-4](https://doi.org/10.1016/S2542-5196(25)00094-4)
- Zhu, H., Martin, R. V., Van Donkelaar, A., Hammer, M. S., Li, C., Meng, J., Oxford, C. R., **Liu, X.**, ... & Lyapustin, A. (2024). Importance of aerosol composition and aerosol vertical profiles in global spatial variation in the relationship between PM_{2.5} and aerosol optical depth. *Atmospheric Chemistry and Physics*, 24(20), 11565-11584. <https://doi.org/10.5194/acp-24-11565-2024>
- Perez-Plazola, M., Diaz, J., Stewart, C. P., Arnold, C. D., Caswell, B. L., Lutter, C. K., Werner, E. R., Maleta, K., Turner, J., Prathibha, P., **Liu, X.**, Gyimah, E., Iannotti, L. (2023). Plasma Mineral Status after a Six-Month Intervention Providing One Egg per Day to Young Malawian Children: A Randomized Controlled Trial. *Scientific Reports*, 13(1), 6698. <https://doi.org/10.1038/s41598-023-33114-1>.
- Lin, C., **Liu, X.**, Yang, Q., Wu, H., Liu, F., Zhang, Q., Zhu, A., Liu, Q. (2019). Hydrophobic Side Chains to Enhance Hydroxide Conductivity and Physicochemical Stabilities of Side-Chain-Type Polymer AEMs. *Journal of Membrane Science*, 585, 90–98. <https://doi.org/10.1016/j.memsci.2019.04.066>.

PRESENTATIONS

- American Geophysical Union Fall Meeting (**AGU 2024**), *Assessing Attenuation Effects in X-ray Fluorescence Analysis of Light Elements in Mineral Dust for Ground-Based Measurements*. Washington, D.C. **Poster**.
- American Geophysical Union Fall Meeting (**AGU 2023**), *Elemental Composition of Fine Particulate Matter in Central Asia: Dust and Trace Elements in Tashkent, Uzbekistan*. San Francisco, CA. **Poster**.
- American Association for Aerosol Research Annual Conference (**AAAR 2023**), *Elemental Characterization of Ambient Particulate Matter for a Globally Distributed Monitoring Network: Methodology and Implications*. Portland, OR. **Talk**.
- The 4th International Surface PARTiculate mAtter Network (SPARTAN) Meeting (**2023**), *Advances in elemental characterization with the XRF*. St Louis, MO. **Talk**.
- American Geophysical Union Fall Meeting (**AGU 2022**), *Elemental Characterization of Ambient Particulate Matter for a Globally Distributed Monitoring Network: Methodology and Implications*. Chicago, IL. **Talk**.
- American Association for Aerosol Research Annual Conference (**AAAR 2021**), *A Global-scale Mineral Dust Equation*. Virtual. **Talk**.
- Atmospheric Optics: Aerosols, Visibility, and the Radiative Balance Conference (**2021**), *A Global-scale Mineral Dust Equation*. Bryce Canyon, UT. **Talk**.
- American Geophysical Union Fall Meeting (**AGU 2020**), *A Global-scale Mineral Dust Equation*. Virtual. **Poster**.

SELECTED AWARDS & HONORS

- Second Prize in the 26th annual Graduate Research Symposium, Washington University in St. Louis Graduate Student Senate and Graduate Professional Council, 2021
- Outstanding Graduate, Xiamen University, 2019
- National Scholarship (1/60), Ministry of Education of P. R. China, 2017 & 2018
- National First Prize in the 12th National Undergraduate Chemical Design Competition, National Undergraduate Chemical Design Competition Committee, 2018
- Merit Student, Xiamen University, 2016 – 2019

TEACHING & ADVISING EXPERIENCE

Teaching Assistant at Washington University in St. Louis

Air Quality Engineering with Lab (Fall 2022); Kinetics and Reaction Engineering Principles (Spring 2021); Unit Operations

Laboratory (Fall 2020)

Student Mentor at Washington University in St. Louis

Research mentor for graduate students (Yingxi Lu, Xinying Wang, Yuxuan Ren) and undergraduate students (Caroline DeLuca, Yuanlong Su)

PROFESSIONAL ACTIVITIES & SERVICE

- **Session chair** for AAAR 2023 Annual Conference, 2023
- **Site manager** for 6 sites in the SPARTAN network (Abu Dhabi, UAE; Dhaka, Bangladesh; Mexico City, Mexico; Seoul, South Korea; Sherbrooke, Canada; Ulsan, South Korea), 2022 – 2024
- **Journal reviewer** for Atmospheric Chemistry and Physics, Aerosol Science & Technology

OUTREACH

- **Event coordinator** of the WashU Center for Aerosol Science and Engineering (CASE) Student Chapter Committee, 2023 – 2024
- **Lab representative** at WashU Department of Energy, Environmental and Chemical Engineering Graduate Student Council, 2021 – 2022

TECHNICAL SKILLS

- Familiar with Weather Research and Forecasting Model with Chemistry (WRF-Chem)
- Skilled in elemental characterization techniques, including X-ray Fluorescence (XRF) and Inductively Coupled Plasma Mass Spectrometry (ICP-MS)
- Experiences in field sampling of ambient particulate matter
- Skilled in R, ArcGIS; Familiar with Python, MATLAB, ENVI, ICP-OES